



NORTHERN UNIVERSITY

OF BUSINESS & TECHNOLOGY KHULNA

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Week 07: Frontend Progress Report

MatchPoint AI

AI Resume Analyzer & Interview Preparation Platform

Project Title	MatchPoint AI
Report Type	Week 07 Frontend Development & API Integration Progress
Course Title	Software Development III
Course Code	CSE 4104
Team	CSE4104-7A-T05
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Date	06/07/2026

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1. Project Overview

MatchPoint AI is an AI-powered resume analysis and interview preparation platform. The system allows users to register, log in, upload a resume, submit a target job description, receive an ATS-style match score, identify missing keywords and skills, view improvement suggestions, generate interview questions, and track previous analysis history.

Frontend Progress Objective: The Week 07 development work focused on implementing the approved UI/UX design as a working React frontend, connecting it with backend APIs, enabling authentication, validating user inputs, and preparing a responsive user interface for demonstration and submission.

Primary User Roles

Role	Frontend Responsibility
Candidate / User	Register/login, upload resume, submit job description, view ATS result, practice interview questions, view profile and history.
Administrator	Monitor users, view analytics, check AI usage information, and review admin logs.

2. Frontend Technology Stack

The frontend has been implemented using the planned React.js, Tailwind CSS, and ShadCN UI stack. Supporting libraries are used for routing, API communication, icons, and Vite-based development/build tooling.

Technology	Purpose in Frontend
React.js	Core frontend library used to build reusable page and layout components.
Tailwind CSS	Utility-first CSS framework used for spacing, colors, typography, layout, responsiveness, and overall UI styling.
ShadCN UI	Component system used for cards, buttons, inputs, labels, textareas, alerts, badges, tables, tabs, progress, skeletons, dropdowns, sheets, and avatars.
Vite	Frontend development server and production build tool.
React Router DOM	Handles navigation between public, candidate, and admin routes.
Axios	Connects frontend components with backend REST APIs.

Lucide React	Provides clean and consistent iconography across the application.
Build Status: The frontend production build was tested successfully after the Tailwind CSS and ShadCN UI migration.	

3. Backend & Database Context Used for Integration

The frontend was integrated with the backend architecture prepared for MatchPoint AI. The backend uses Node.js and Express.js with Supabase cloud services for database storage and user authentication. Resume parsing and analysis-related data are stored in Supabase PostgreSQL tables.

Backend Area	Implementation Summary
Runtime / Framework	Node.js (v18+) and Express.js (v4.19+)
Database & Auth Engine	Supabase PostgreSQL and Supabase Authentication
Middleware	cors, express.json(), multer memory buffer for resume upload
Document Extraction	pdf-parse for PDF and mammoth for DOCX
Developer Tooling	nodemon for live backend reloads

Database Tables Relevant to the Frontend

Table	Frontend Usage
users	Stores public profile records linked with Supabase Authentication users.
resumes	Stores uploaded resume file information and extracted/parsed content reference.
job_descriptions	Stores job title, company, and job description text submitted by users.
analysis_records	Stores ATS score, missing keywords, missing skills, and analysis results.
interview_sessions	Stores generated mock interview questions linked to analysis data.

4. Completed Frontend Pages

All required public, candidate, AI interaction, and admin monitoring pages have been implemented and visually checked after the Tailwind CSS and ShadCN UI migration.

Page	Route	Completion Summary
Home / Landing Page	/	Introduces MatchPoint AI with CTA buttons and platform feature preview.
Login Page	/login	Allows registered users to sign in using email and password.
Registration Page	/register	Allows new users to create a candidate account.
Dashboard	/dashboard	Shows ATS score, total analyses, saved jobs, interview practice, and recent analyses.
Analyze Resume	/analyze	Allows resume upload, job information input, and AI analysis request.
AI Result	/result	Displays ATS match score, missing keywords/skills, and improvement suggestions.
Interview	/interview	Displays interview practice questions based on analysis/gaps.
Jobs	/jobs	Displays job recommendations and matching scores.
History	/history	Shows previous analysis records for the candidate.
Profile	/profile	Displays and updates candidate profile information.
Admin Panel	/admin and sub-routes	Provides admin dashboard, users, analytics, AI usage, and logs pages.

5. Backend APIs Integrated with Frontend

The frontend communicates with the backend through structured REST API calls. Authentication tokens are stored and attached to protected API requests using the frontend API service configuration.

Method	Endpoint	Frontend Use
POST	/api/auth/register	Registration form creates a new user account and profile.
POST	/api/auth/login	Login form authenticates the user and receives session data.
POST	/api/auth/logout	Logout button clears the active session and redirects safely.
GET	/api/user/profile	Profile page fetches the logged-in user profile.
PUT	/api/user/profile	Profile page saves edited candidate information.
POST	/api/upload	Analyze page uploads resume file and sends extracted/upload data to backend.
POST	/api/analysis/gap-analysis	Analyze page submits resume and job description data for AI gap analysis.
GET	/api/user/history	Dashboard and history page show previous analysis records.

6. Authentication & Protected Route Flow

The authentication workflow is connected with Supabase Authentication through the backend API. The frontend stores session data after login, protects private routes, and displays unauthorized users through redirect handling.

Step	Flow Area	Implementation Summary
1	Registration	User submits full name, email, and password through the registration page.
2	Supabase Account Creation	Backend creates the user through Supabase Auth and syncs the public profile row.

3	Login	User signs in through the login page and receives a valid session/JWT from the backend.
4	Protected Access	Private routes such as dashboard, analyze, history, and profile are accessible only after login.
5	Logout	Visible logout button clears local session data and returns the user to the public/auth flow.

Verification Result: Register, login, logout, protected route access, dashboard redirect, and Supabase user/profile table updates were tested successfully during frontend integration.

7. Resume Analysis Workflow

The major functional workflow connects file upload, job description input, backend parsing, AI gap analysis, database persistence, and result rendering into one complete user journey.

Workflow	Description
Step 1	Candidate opens the Analyze Resume page.
Step 2	Candidate uploads a PDF/DOCX resume and enters job title, company, and job description.
Step 3	Frontend validates the fields and submits data to backend upload and analysis APIs.
Step 4	Backend extracts text, performs resume-versus-job gap analysis, and stores records in Supabase.
Step 5	Frontend redirects to AI Result page showing ATS score, missing keywords, missing skills, and suggestions.

8. Responsive Design & UI/UX Improvements

The frontend UI follows a modern SaaS dashboard layout with a navy sidebar, white cards, blue and purple accent colors, clean forms, and responsive behavior for desktop and mobile screen sizes.

UI/UX Area	Implementation
Responsive Layout	Desktop sidebar, responsive grids, mobile menu/sheet navigation, and readable mobile cards.
Input Validation	Required fields, email validation, password confirmation, file type/size checks, and readable errors.
Loading Feedback	Loading states and disabled buttons during form/API actions.
User Feedback	Friendly error alerts, success actions, clean navigation, and empty-state handling.
Visual Consistency	Tailwind utility classes and ShadCN UI components applied across forms, cards, buttons, tables, and dashboards.

9. Testing and Verification Summary

The following checks were completed after final frontend migration and QA fixes.

Test Area	Result
Production Build	npm run build completed successfully with no build errors.
Login Form	Submit button and login handler verified.
Registration Form	Submit button and registration handler verified.
Analyze Form	Analyze with AI submit action verified and result page displayed.
Logout	Visible desktop and mobile sidebar logout verified.
Supabase Integration	Authentication users and database table rows verified in Supabase.
Responsive View	Mobile responsive view checked through Chrome device toolbar.

10. Current Development Progress

Week 07 frontend development is functionally complete. The frontend now includes all required pages, authentication flow, backend API integration, resume analysis workflow, Tailwind CSS and ShadCN UI implementation, responsive layout, validation, and submission screenshots.

Requirement Area	Current Status
Frontend Setup	Completed
Required Pages	Completed
Tailwind CSS + ShadCN UI Migration	Completed
Backend API Integration	Completed
Supabase Authentication	Completed
Supabase PostgreSQL Data Persistence	Completed
Responsive UI	Completed
Screenshots	Screenshots prepared and GitHub repository screenshot added after final push.
Documentation	Prepared in pdf format

11. Screenshot Evidence

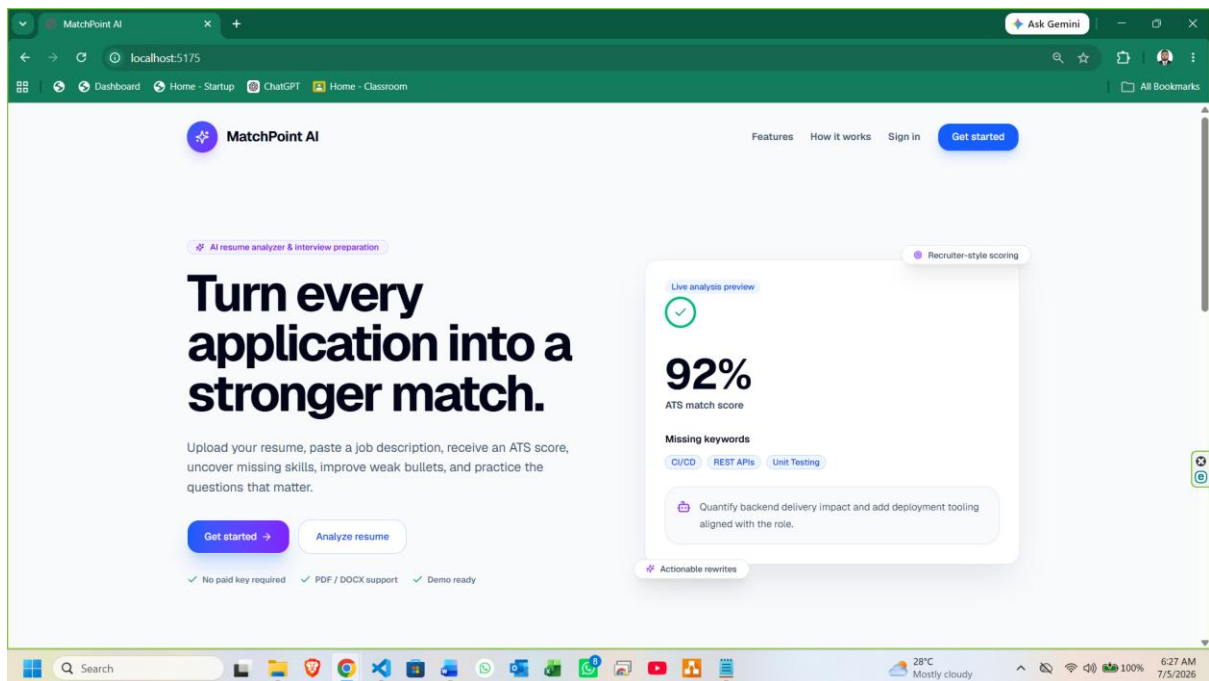


Figure 1: Home Page of MatchPoint AI

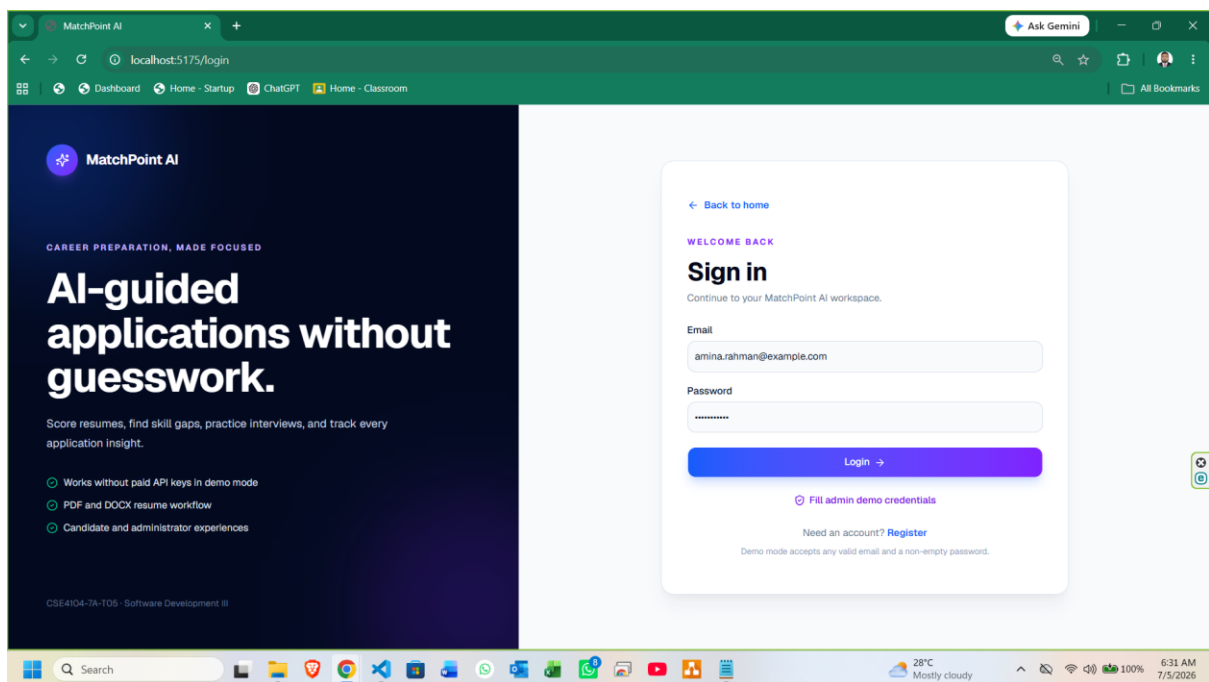


Figure 2: Login Page with frontend authentication form

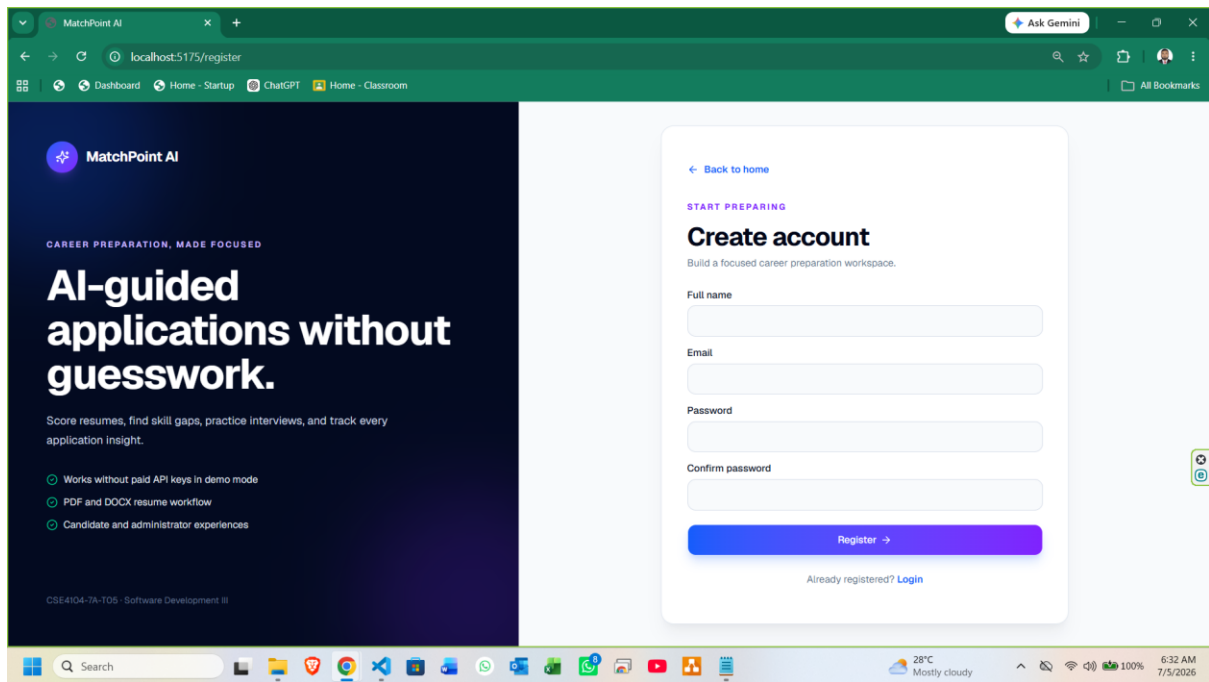


Figure 3: Registration Page with account creation form

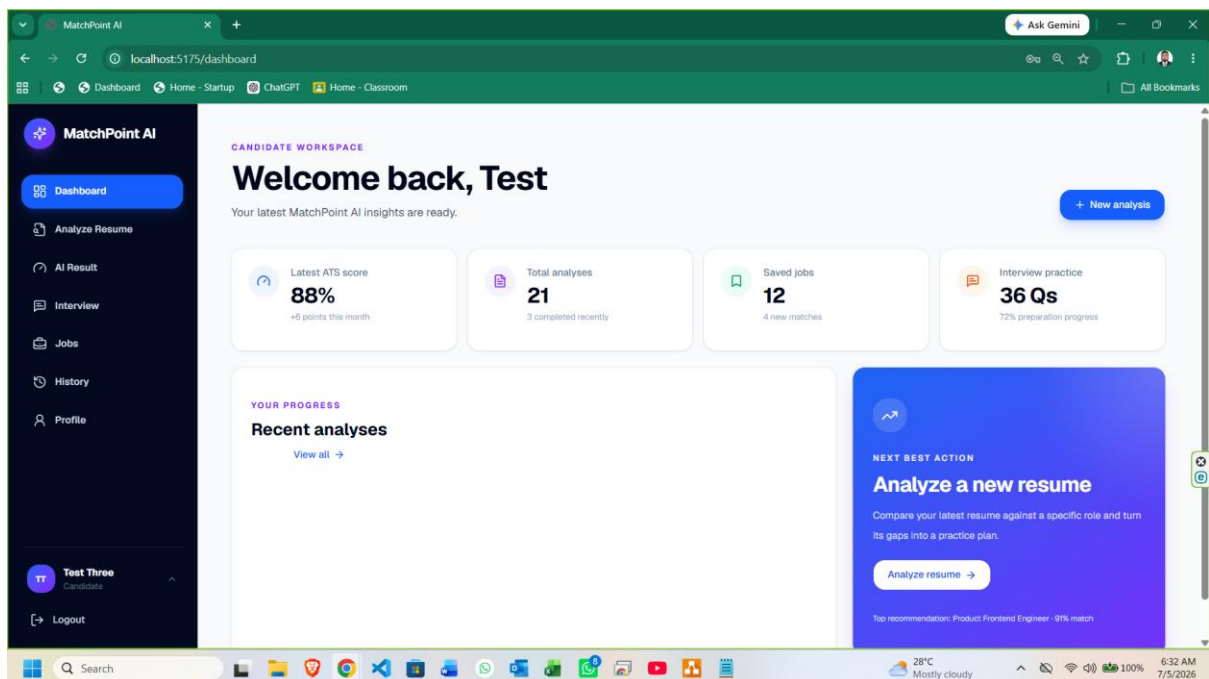


Figure 4: Candidate Dashboard after successful login

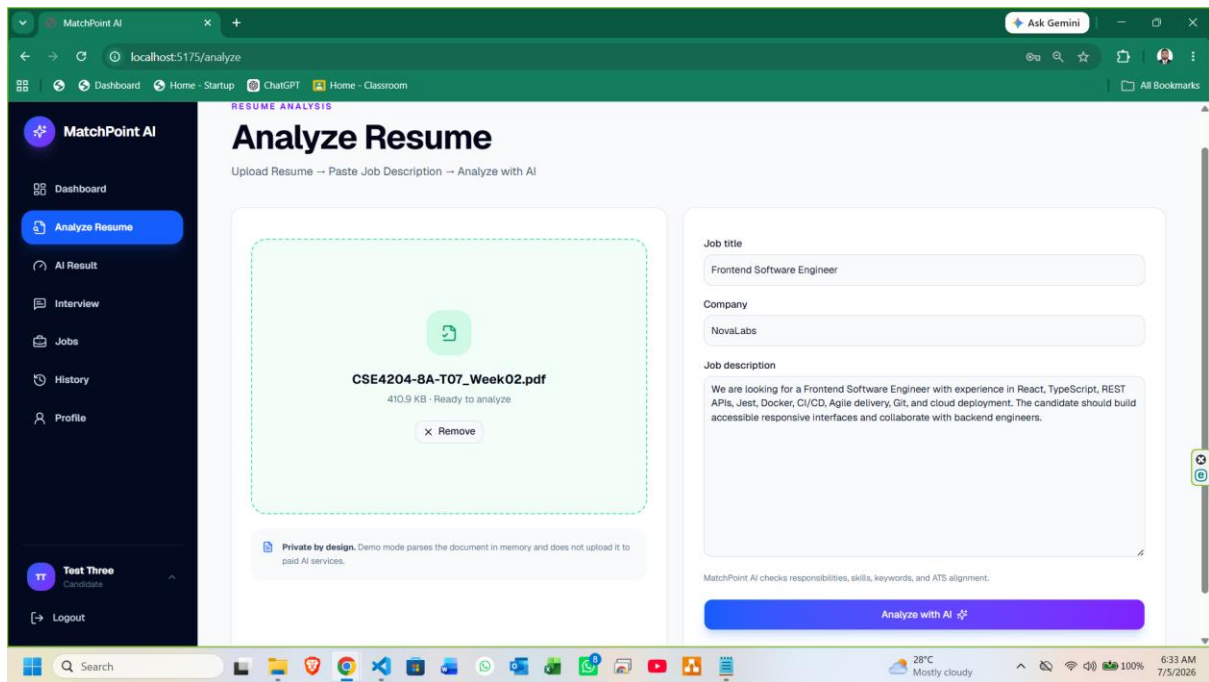


Figure 5: Analyze Resume page with upload and job description input

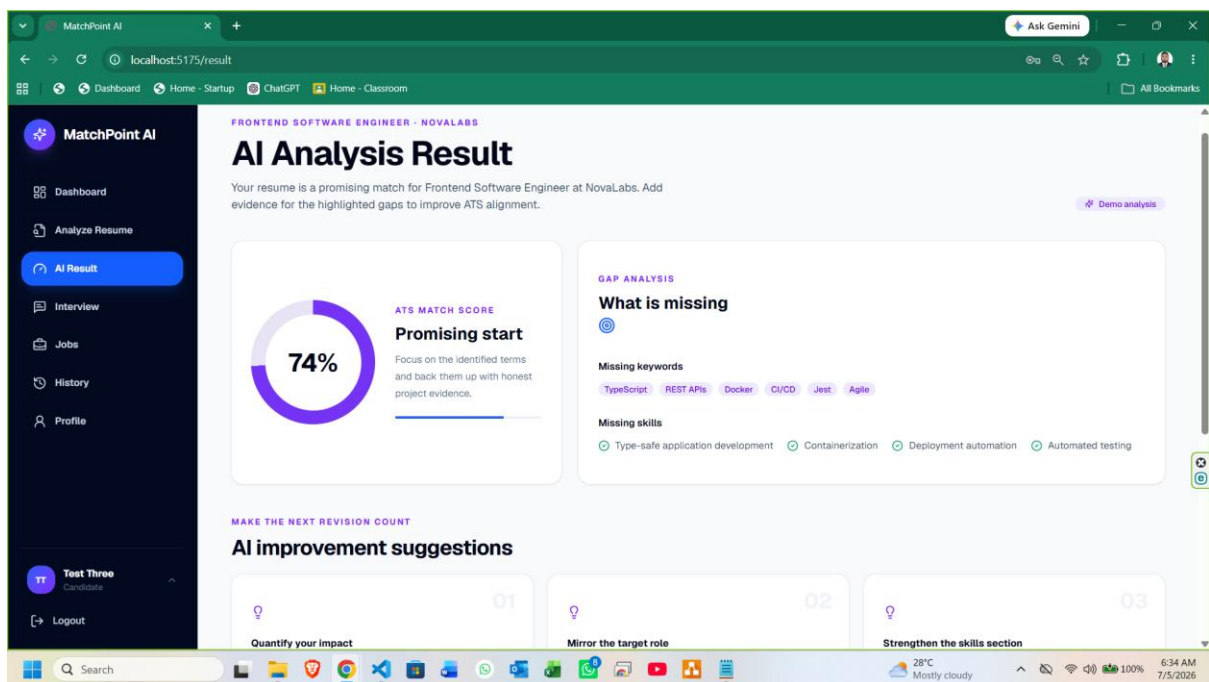


Figure 6: AI Analysis Result page with ATS score and gap analysis

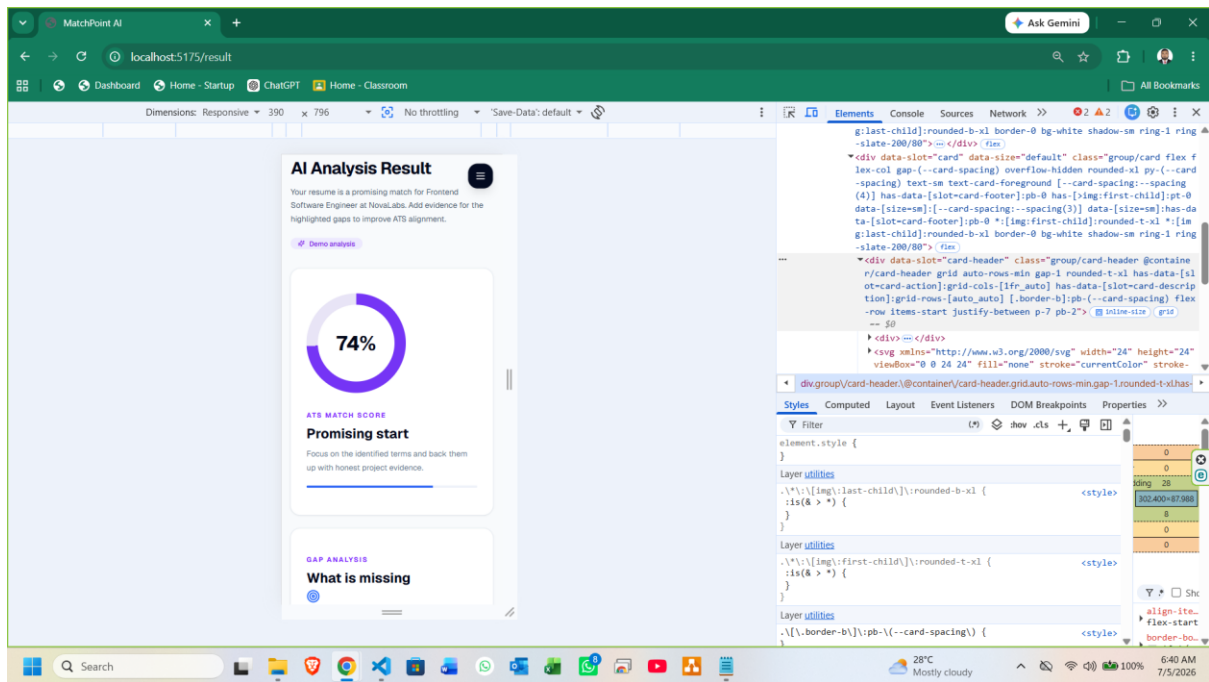


Figure 7: Responsive Mobile View using Chrome device toolbar

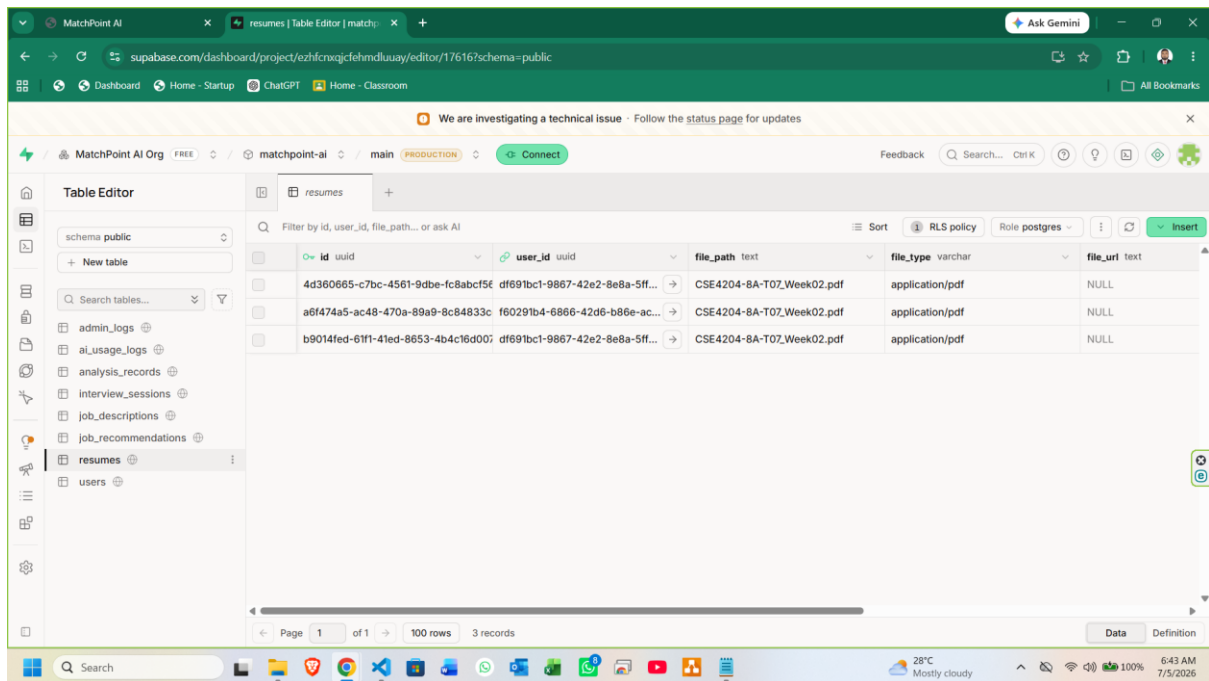


Figure 8: API Integration proof from Supabase table records

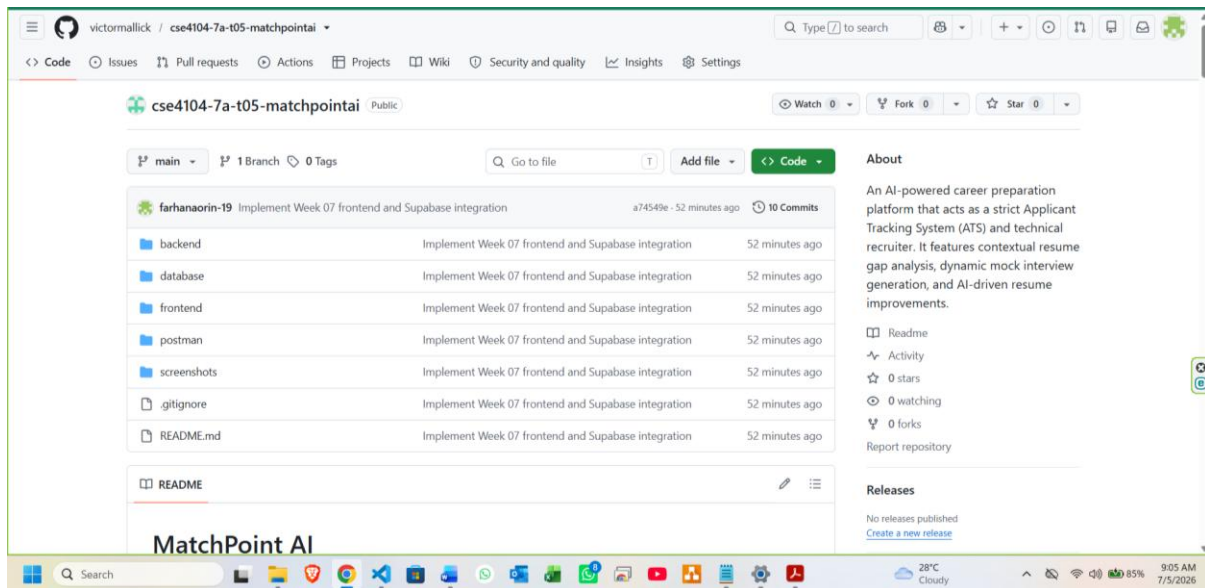


Figure 9: GitHub Repository after final push

12. GitHub Repository & Submission Checklist

Repository Link: <https://github.com/victormallick/cse4104-7a-t05-matchpointai>

Repository Item	Submission Status
frontend/	Contains React.js, Tailwind CSS, ShadCN UI pages and components.
backend/	Contains Node.js and Express.js backend API logic.
database/	Contains Supabase/PostgreSQL schema and SQL setup files.
documentation/	Contains this Week 07 Frontend Progress Report.
screenshots/	Contains required screenshot evidence for submission.
README.md	Project overview, setup instructions, stack, and API summary.
.env files	Ignored and not pushed to GitHub.